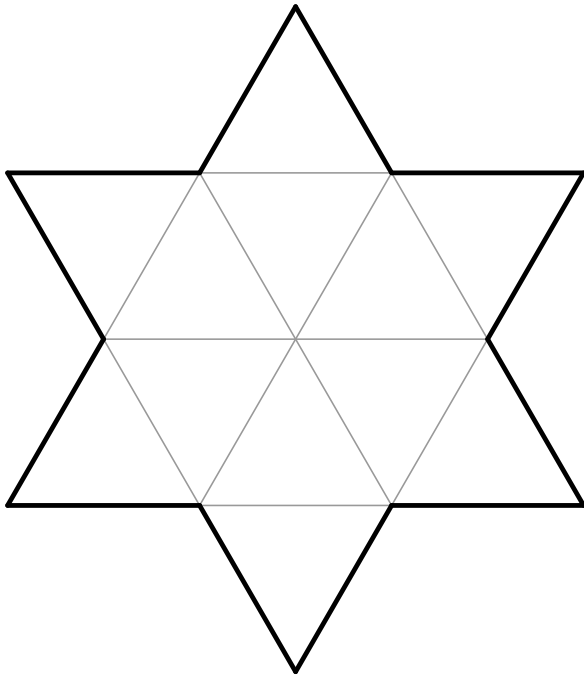
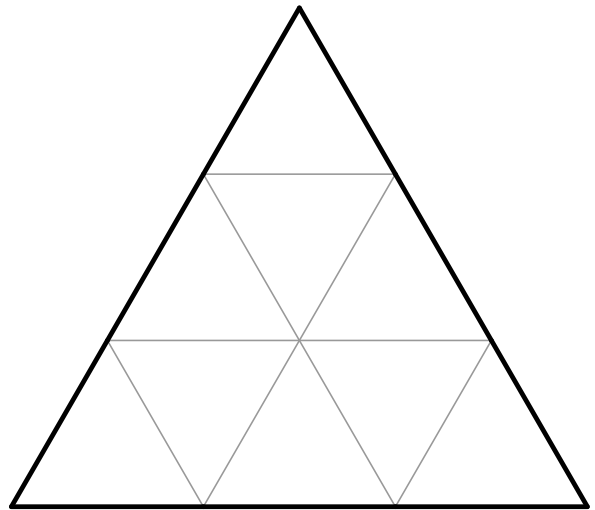


On every board in this packet, each block must cover whole small triangles of the board. Blocks may not overlap or stick out past the thick outline. Use only the small blocks.

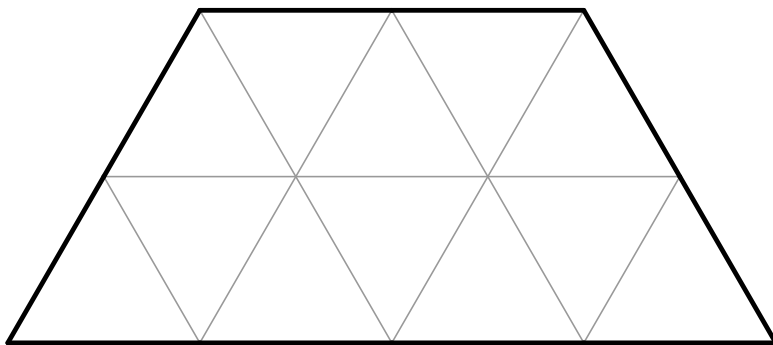
Problem 1: Cover each board with blue rhombuses , or decide that it cannot be done. For each board that cannot be covered, explain why.



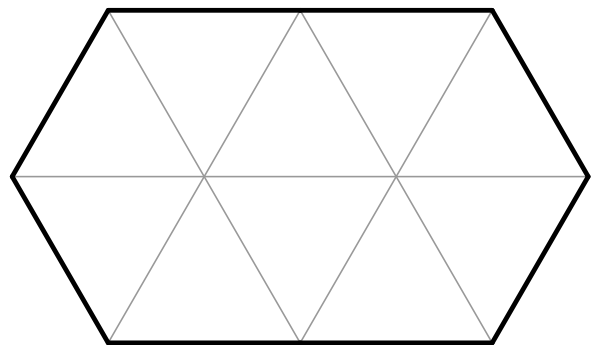
A



B

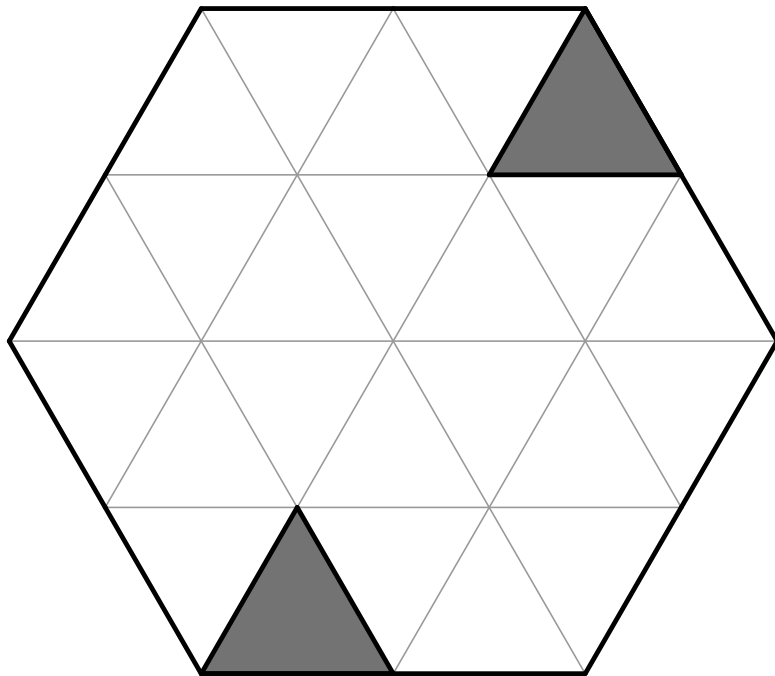


C

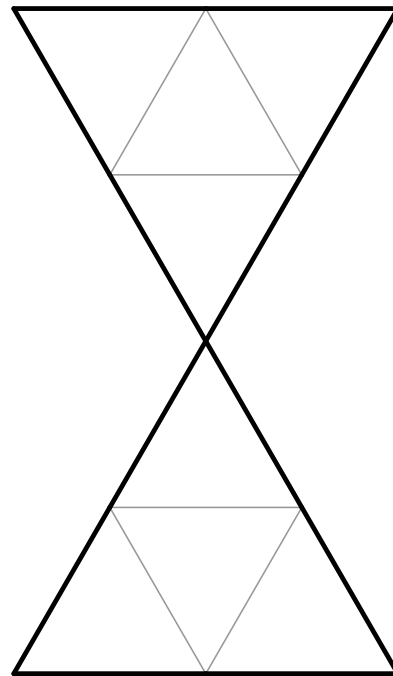


D

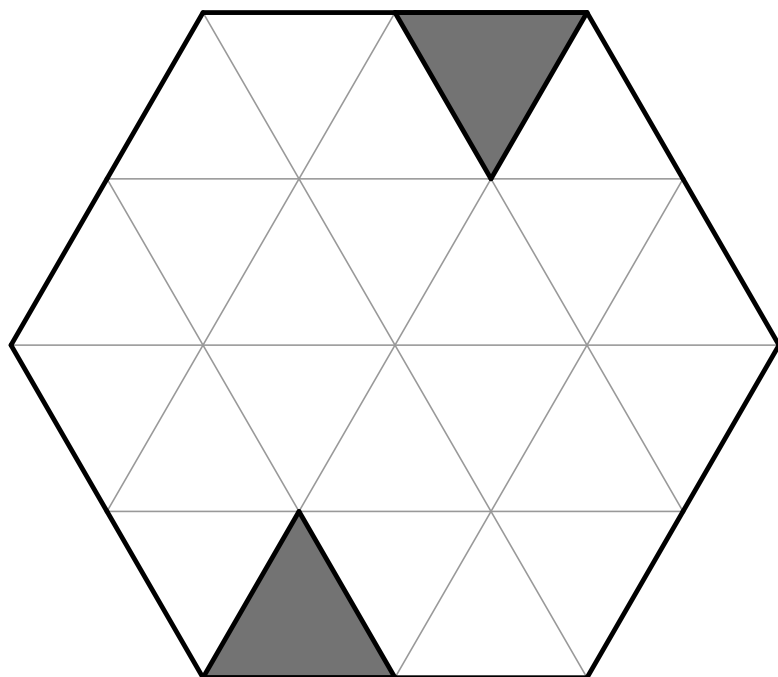
Problem 2: The gray triangles are holes, so they are not part of the board. Cover each board with blue rhombuses , or decide that it cannot be done. For each board that cannot be covered, explain why.



E

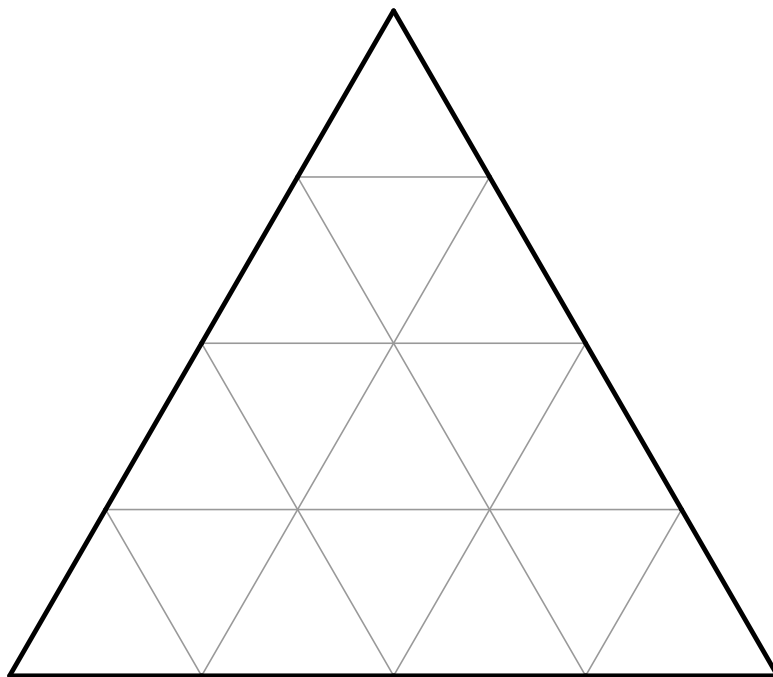
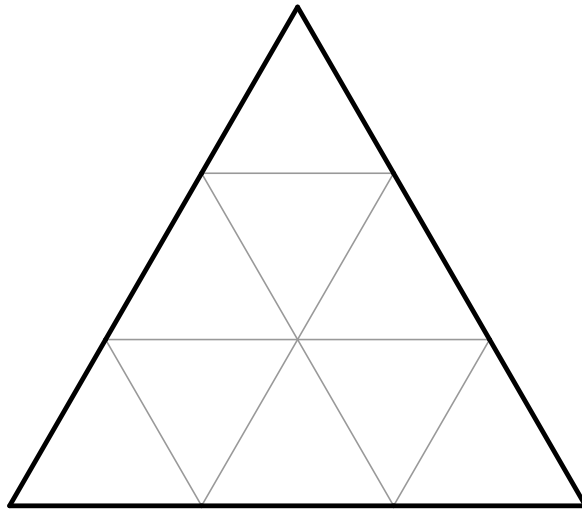
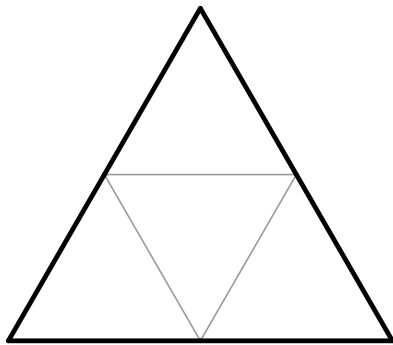


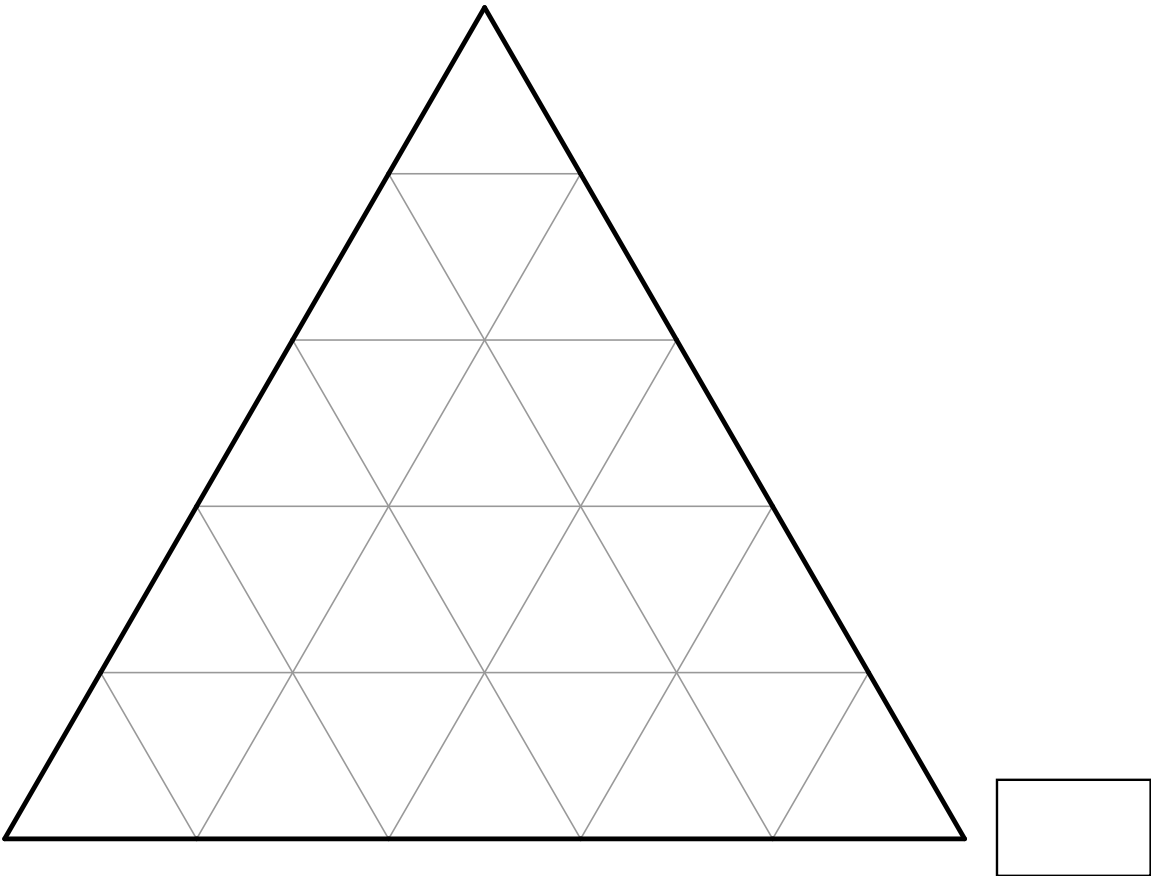
F



G

Problem 3: Cover each big triangle on this page and the next with blue rhombuses  and green triangles . Use as few green triangles as you can, and write that number in the box next to the triangle. Then explain why nobody can cover the biggest triangle with fewer green triangles than you used.



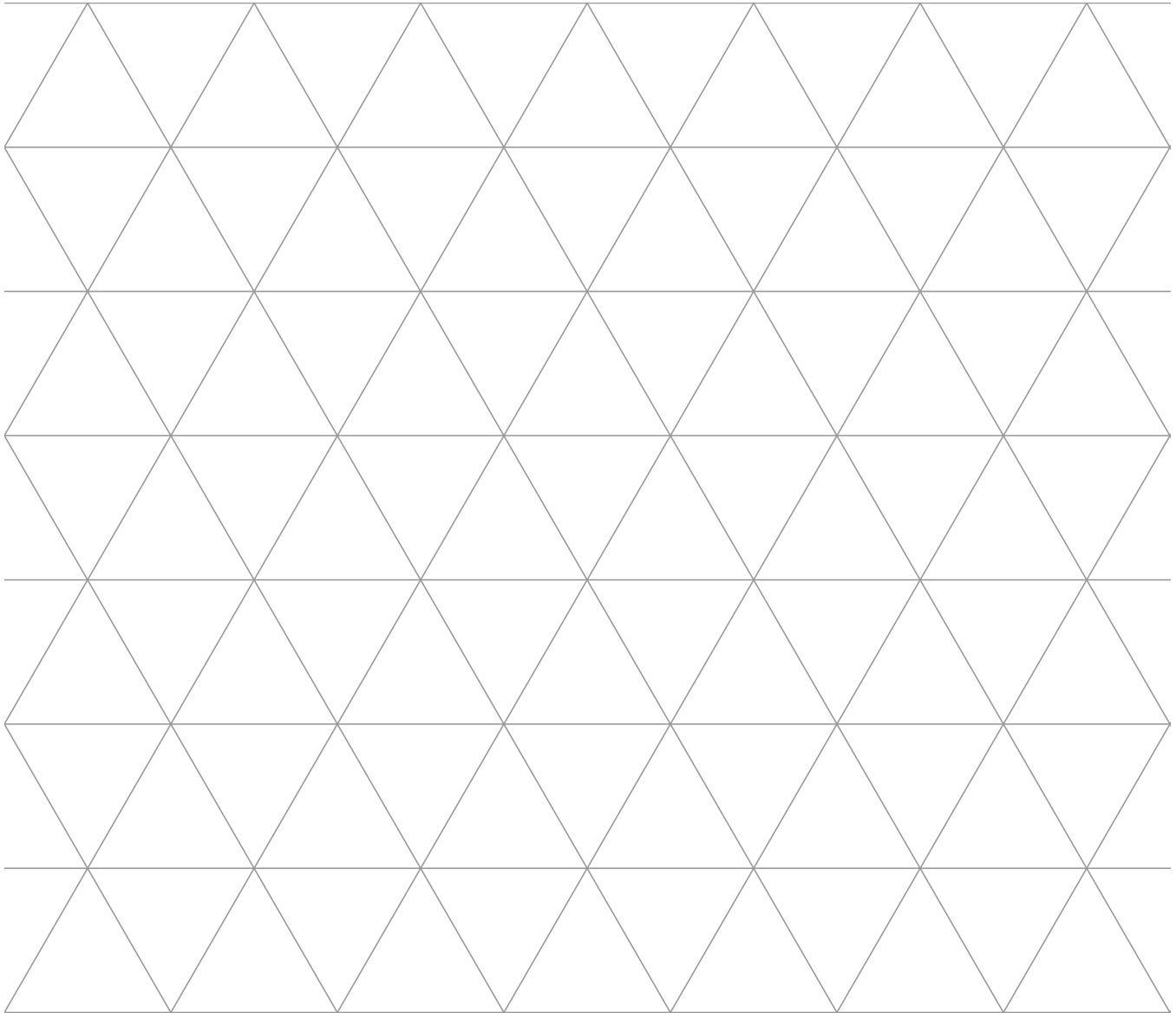


Problem 4: Imagine a triangle board with 10 small triangles along each side. If you cover it with blue rhombuses and green triangles, what is the smallest number of green triangles you need? Explain why nobody can use fewer. What if the board has 100 small triangles along each side?

Problem 5: Go back to the four triangles in Problem 3. This time cover each one with purple chevrons  and green triangles  , using as few green triangles as you can. Write your numbers in the table. Can purple chevrons ever need fewer green triangles than blue rhombuses? Explain.

	with blue rhombuses	with purple chevrons
2 small triangles on each side		
3 small triangles on each side		
4 small triangles on each side		
5 small triangles on each side		

Problem 6: On the grid below, draw the outline of a board made of exactly 20 small triangles that needs at least 6 green triangles when you cover it with blue rhombuses and green triangles. The board must stay in one piece if you cut it out.



Problem 7: Put two green triangles  anywhere on this hexagon board. Can blue rhombuses  cover the rest of the board? Find a rule that tells which places for the two green triangles work, and explain why the other places can never work. You can mark the places you try on the small hexagons.

